

ABOUT US

Gravity was founded in South Africa in 1999. Since then Gravity has been a leader in the work at height industry. Assisting in developing a variety of standards, providing excellent training, producing innovative solutions for work sites as well as a range of fall arrest equipment.

Gravity is committed to providing quality services and products with our clients' satisfaction being our highest priority.

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INTRODUCTION

Gravity is a global leader in permanent fall protection for a range of applications and scenarios. Gravity has partnered with FallProtec to manufacture rope and rail permanent lifelines.

Permanent systems require training specialized for installation and use.



PURPOSE

This booklet is a training aid that allows the learner, with the guidance of a competent facilitator, to understand the requirements, limitations and safe use of Gravity systems.

SCOPE

Only the safe use of Gravity Vertical and Horizontal Systems will be discussed in this document.

It does NOT extend to installations or detailed inspections of any systems nor does this include the use of other permanent fall protection systems.

Should you encounter other Gravity systems, ensure that you receive the relevant training and ensure you follow the manufacturer's requirements.

BEFORE WORK STARTS

- At least 2 technicians on site
- Conduct full risk assessment
- Ensure all workers are medically fit
- All equipment must be inspected before work starts
- Have a rescue plan with a rescue kit in place



Before any Gravity systems may be used the site supervisor must ensure that the following is in place:

- Must be installed by a competent person
- Must be inspected once a year
- Must be signed off
- User must be trained and practiced in use thereof as well as fall arrest

ALL OF THE INFORMATION MUST BE ENGRAVED ON THE CERTIFICATE PLATE.

Scan QR code



Manufactured by:

FALLPROTEC

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F: +352 26 55 09 30 55
www.fallprotec.com

Installed by:

GRAVITY ACCESS

GRAVITY ACCESS
Unit 24, Scientia Technopark, Meiring Neude road,
Broomfield
Pretoria
South Africa
0861101213
www.gravitygh.co.za

EN795-C:2012
13th April 2015

Type of system: SecuRope 2012 horizontal/inclined

BEYOND THIS AREA WEARING A
HARNES AND SECURED BY A FALL
ARREST SYSTEM ARE MANDATORY

Installation date: 2020-08-28

Company: MBSA

Maximum number of users on line: 4

Location: MBSA

Length: 258 m

Paintline: MBSA Paintline Life-line

Inspection date: 2020-08-28

Type of system

Installation date

Maximum number of
users on line

Inspection date

GRAVITY VERTICAL SYSTEM

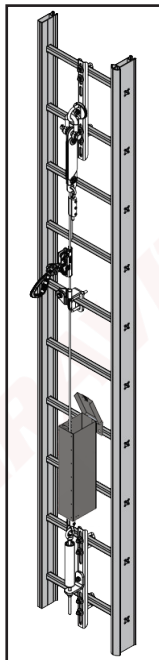
The Gravity Vertical System is constructed with a 8mm 7x7 stainless steel wire rope running through intermediate anchors, connected to end anchors at both ends usually with a tensioning device at one end at the bottom of the line, complemented with a storage box for multiple Gliders.

ADVANTAGES AND DISADVANTAGES

This line offers some advantages over other fall protection systems. It reduces the fall risk as well as the fall factor a worker is exposed to. It also allows for more efficient work on-site as it allows for faster ascent and descent than would be possible with shock absorbing lanyards. This system also allows multiple Gliders / users to be connected to it.

The Gravity Vertical System incorporates an intelligent cable vibration absorption element. This allows the life line to move with the tower when exposed to side forces such as wind and expansion and contraction due to temperature changes.

However, the system does restrain the worker to the line and should the worker be required to move away from the installed system, shock absorbing lanyards must be taken up as well.



EQUIPMENT



EN361 - Full
Body Harness



EN353-1 - Glider Fall
Arrester with rigid line



EN355 - Shock
Absorber



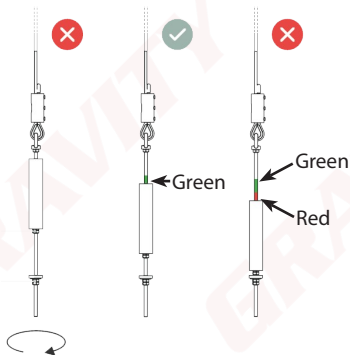
EN358 - Workbelt



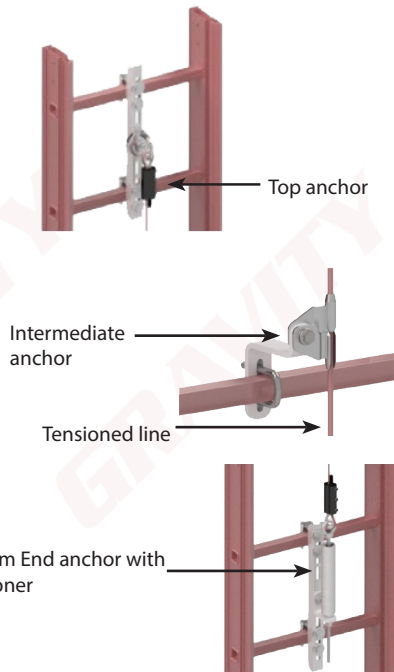
EN12492 - Helmet
GLCGLSLM_Rev 2.3

PRE-USE CHECKS

Before a user connects to a Gravity Vertical System, the following elements of the system must be inspected and determined to be safe for use.



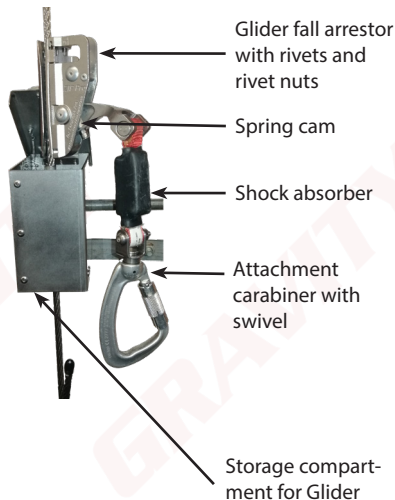
When tensioning the lifeline, take the colour code into account. Green indicates proper tension, red too high tension and no colour insufficient tension.



PRE-USE CHECK OF THE GLIDER

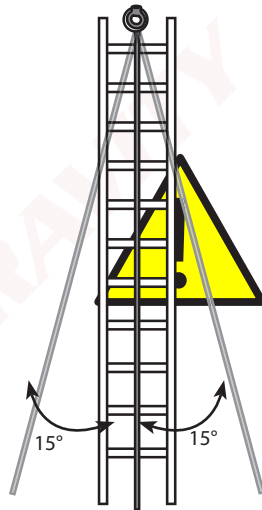
After the Pre-use check of the system has been completed the user needs to ensure that the Glider forms part of the Gravity Vertical system.

- The Glider fall arrester needs to be compatible with the system.
- Do a function test on ground level by lifting the cam arm and letting go, the arm should spring back and the cam will engage on the system.
- Inspect the shock-absorber for deployment and report it to the supervisor if it is damaged.
- Inspect all the other components of the Glider fall arrester for any defects and or damages before use. Such as : Carabiner with swivel, Glider fall arrester, rivets , rivet nuts and spring cam.



USE AND LIMITATIONS

- The number of users allowed on the line is given on the certificate plate, but will not exceed more than 4 users for this system
- In the event of multiple users on the system it is best to maintain a minimum safe distance of 6 meters whilst climbing, should you need to work closer to each other the top user needs to apply his work belt or remove himself from the system by connecting an alternative safety to the structure and connect the Glider to one of the ladder rungs
- 15° maximum inclined angle
- 50kg-120kg users only
- Only use the Glider fall arrester
- Keep your body close to the system to allow for better movement
- 10m max length between supporting anchors
- 8mm stainless steel wire rope
- All components must be of compatible material
- Do not use the system as a work positioning
- Do not stand or step on any of the components of the system
- Do not use in inclement weather



WARNINGS

- Never remove the Glider from the line before additional fall protection has been added while at height
- Do not manipulate the fall arrester directly
- Do not use the Glider fall arrester if its shock absorber if it is longer than 12cm
- Do not use if the line is slack
- Do not ground a welding machine on a line
- Never use the line outside of its limits
- Do not use if the inspection is not current
- Do not adjust, alter or tamper with the system in any way.
- Do not extend the Glider fall arrester's shock absorber.



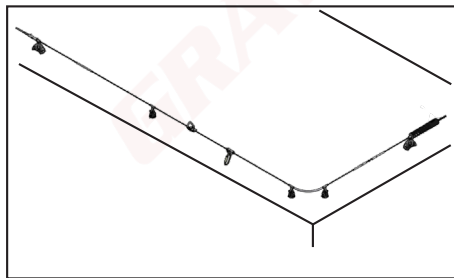
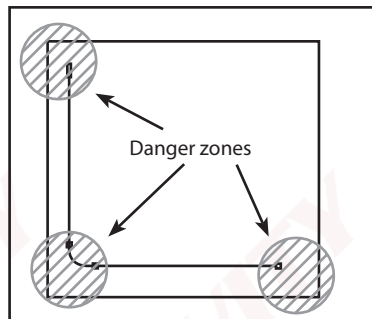
GRAVITY HORIZONTAL SYSTEM

The Gravity Horizontal System is constructed with a 8mm 7x7 stainless steel wire rope running through intermediate anchors, connected to end anchors at both ends usually with a tensioning device at one end. The Horizontal line also includes a shock absorber which reduces the chance of injury in the case of a fall as well as the force on system/structure.

ADVANTAGES AND DISADVANTAGES

This line offers some advantages over other fall protection systems. It reduces the fall risk as well as the fall factor a worker is exposed to. It also allows for more efficient work on-site as it allows for faster movement than would be possible with shock absorbing lanyards and gives access to areas where there are no other anchors available.

However, the system does restrain the worker to the line and should the worker be required to move away from the installed system, shock absorbing lanyards must be taken up as well.



EQUIPMENT



EN361 - Full
Body Harness



EN 795-C - Glider

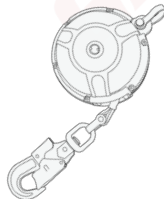


EN355 - Shock
Absorber*

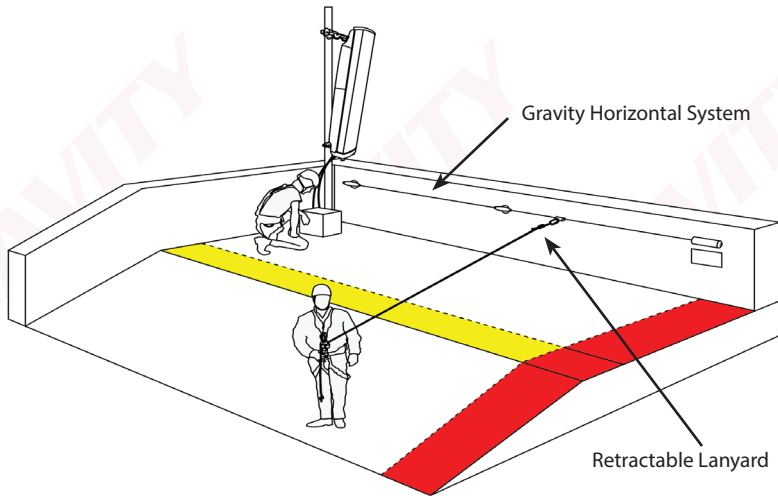


EN12492 - Helmet

*An EN 360 certified retractable lanyards may also be used

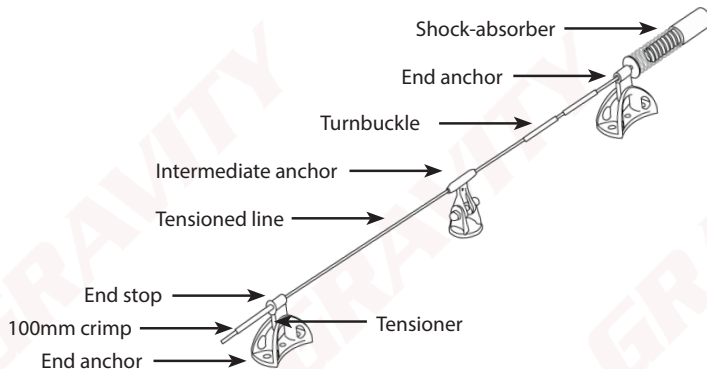


USE OF RETRACTABLE LANYARD ON A GRAVITY HORIZONTAL SYSTEM

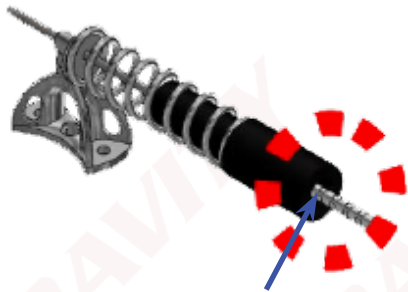


PRE-USE CHECKS

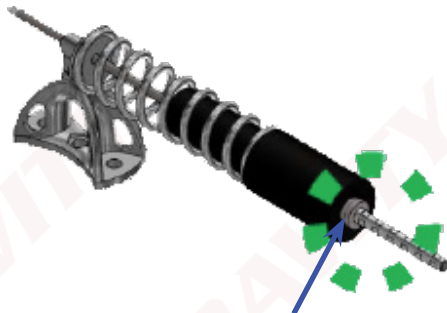
Before a user connects to a Gravity Horizontal System, the following elements of the system must be inspected and determined to be safe for use.



FALL INDICATOR



Fall indicator activated = UNSAFE
Nut not visible

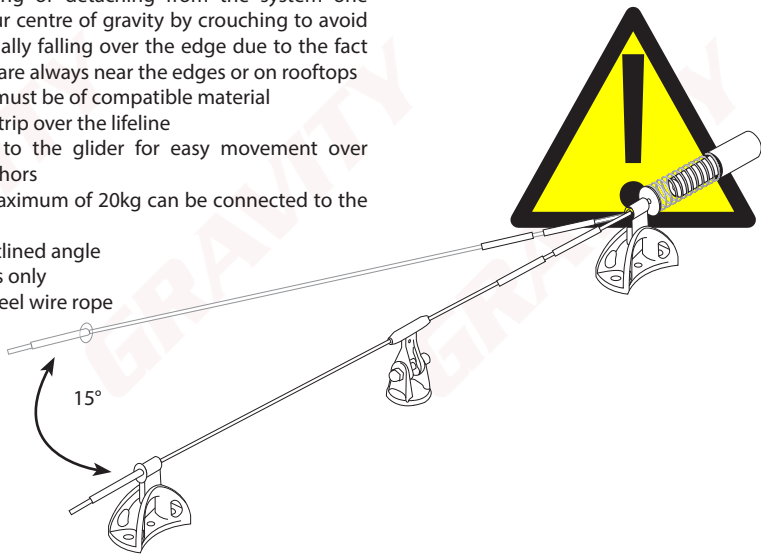


Fall indicator NOT activated = SAFE
Nut visible



USE AND LIMITATIONS

- The amount of users allowed on the line is given on the certificate plate
- When approaching or detaching from the system one should lower your centre of gravity by crouching to avoid you from potentially falling over the edge due to the fact that the systems are always near the edges or on rooftops
- All components must be of compatible material
- Be careful not to trip over the lifeline
- Always connect to the glider for easy movement over intermediate anchors
- Tool bags of a maximum of 20kg can be connected to the lifeline
- 15° maximum inclined angle
- 50kg-120kg users only
- 8mm stainless steel wire rope



WARNINGS

- Never remove the Captive Glider from the line before additional fall protection has been added
- Do not use if the fall indicator on the shock absorber has been activated
- Do not use if the line is slack
- Do not ground a welding machine on a line
- Never use a line outside of its limits
- Do not use if the inspection is not current
- Do not use as rope access anchors
- Do not use any lifeline during an electrical storm
- Never use any overhead lifeline as an anchor for a lifting system



Damaged cable

